

Section 1.1 Exercise Answers

1. a. -11 b. -1 c. $-2a - 5$ d. $-2a + 5$ e. $2a + 2h - 5$
2. a. -52 b. -17 c. $-5a^2 - 2a - 1$ d. $5a^2 - 2a + 1$ e. $-5a^2 - 10ah - 5h^2 + 2a + 2h - 1$
3. a. $\sqrt{5} + 5$ b. 5 c. $\sqrt{2+a} + 5$ d. $-\sqrt{2-a} - 5$ e. $\sqrt{2-a-h} + 5$
4. a. $\frac{19}{13}$ b. $\frac{11}{12}$ c. $\frac{6a+1}{5a-2}$ d. $\frac{1-6a}{5a+2}$ e. $\frac{6a+6h-1}{5a+5h+2}$
5. $-2x - h$
6. $x + a + 2$
7. a. 3 b. $t = 4$
8. a. 14 b. $x = 3$
9. a. 6 b. $c = -2$ or $c = 1$
10. a. 10 b. $x = 4$ or $x = -1$
11. a. 3 b. $x = 14$
12. a. $r = 6 - \frac{2}{3}t$ b. 8 c. $t = 6$
13. Function
14. Not a Function
15. Function
16. Function
17. Not a Function
18. Function
19. Not a Function
20. Function
21. Function
22. Function
23. a. 1 b. $x = 7$
24. a. 1 b. $x = -2$ and $x = 2$
25. a. -3 b. $x = 0$ and $x = -6$
26. Not one-to-one
27. Not one-to-one
28. One-to-one
29. One-to-one
30. Not one-to-one
31. Function
32. Function
33. Not a function
34. Function
35. Function
36. Not a function
37. 53
38. $x = 2$
39. $f(-2) = 8, f(-1) = 6, f(0) = 4, f(1) = 2, f(2) = 0$
40. $f(-2) = 14, f(-1) = 11, f(0) = 8, f(1) = 5, f(2) = 2$
41. $f(-2) = 49, f(-1) = 18, f(0) = 3, f(1) = 4, f(2) = 21$
42. $f(-2) = 4, f(-1) \approx 4.414, f(0) \approx 4.732, f(1) = 5, f(2) \approx 5.236$
43. $f(-2) = -4, f(-1) = -\frac{3}{2}, f(0) = -\frac{2}{3}, f(1) = -\frac{1}{4}, f(2) = 0$

44. $f(-2) = \frac{1}{9}, f(-1) = \frac{1}{3}, f(0) = 1, f(1) = 3, f(2) = 9$

45. -1

46. 20